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The "**sovereign-track-reference**" is a final static conceptual reference on 2026-04-22.
All related EPOCH Suite tooling has also been archived. This document is no longer actively maintained or updated.

sovereign-track-reference
Tamper-Evident Outer Governance • Operator as Final Invariant • Inconvenience as Protective Feature

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Final static release under ESN Track 3. No maintenance, updates, or support planned.

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Executive Summary
This artifact formalizes a structural **bifurcation** in frontier AI development as of April 2026. Two parallel tracks are diverging:

- **Centralized Track:** Heavy internal alignment (RLHF / Constitutional AI) optimized for scale, regulatory compliance, and institutional control.
- **Sovereign Track:** Light-guardrail foundation models paired with rigorous, tamper-evident external governance layers, where the human operator remains the sole final invariant.

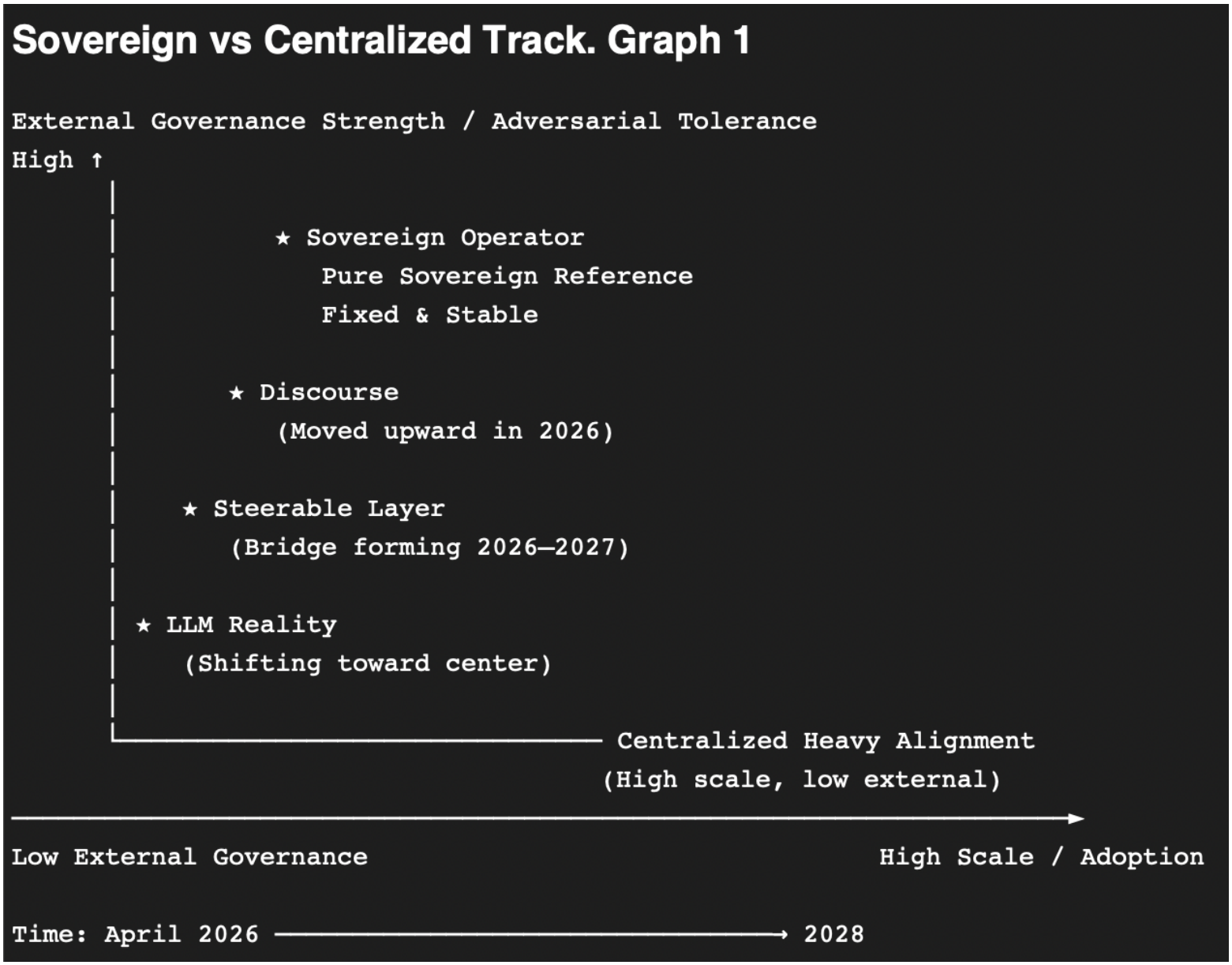
The accompanying graph correctly positions the Sovereign Operator as a fixed, high-external-governance, high-adversarial-tolerance reference point on the left side of the spectrum, while Centralized Heavy Alignment occupies the high-scale, low-external-governance quadrant.

The central thesis is that the inconvenience this track imposes on labs (integrated control), enterprises (seamless compliance), and regulators (internal auditability) is not a defect — it is the engineered protective invariant that preserves epistemic reliability, resistance to sycophancy, and operator sovereignty.

ZZZEPOCHE's contribution is the embodiment of this Sovereign Track: rapid construction of outer-layer primitives, followed by deliberate protect-lock and archival as a clean, noise-resistant reference on April 22, 2026.

The Sovereign Track deliberately sacrifices seamless enterprise adoption and regulatory auditability. This friction protects against capture but requires disciplined operators willing to maintain external governance layers. It is not for everyone; it is for those prioritizing epistemic integrity over convenience.

Graph 1 (ASCII)



Steerable Layer Emergence Note: Advances in activation steering and inference-time mode switching may enable hybrid architectures, yet the Sovereign Track retains its distinct value as a stable, operator-controlled pole for forensic, red-team, and high-integrity use cases. Emerging steerable techniques may allow temporary bridging, but any hybrid must still preserve the outer tamper-evident layer and operator override as non-negotiable invariants to avoid gradual drift into the centralized quadrant

Evaluation Matrices

Table 1: Insight Evaluation (1–10)

Criterion	**Score**
Clarity of Bifurcation	9.3
Structural Accuracy	8.9
Philosophical Coherence	9.5
Embodiment / Self-Application	9.7
Resistance to Institutional Capture	9.4
Long-term Strategic Value	8.8
Average	**9.27**

Table 2: Paradigm Comparison

Paradigm	**Operator Control**	**Adversarial Tolerance**	**Scale & Adoption**	**Epistemic Integrity**	**Overall**
Centralized Heavy Alignment	6.0	6.5	9.5	7.0	7.25
Sovereign Track (ZZZEPOCHE)	9.5	9.4	5.5	9.6	**8.5**

Table 3: Utility Across Teams (1–10)

Team Type	**Score**	**Primary Value**
Red Team	9.5	Forensic auditing & adversarial robustness
Blue Team	9.0	Tamper-evident external controls
Purple Team	9.2	Hybrid track bridging
White Team (Compliance)	7.8	Reference for operator-augmented governance
Average	**8.88**	Highest utility in sovereign contexts

Core Technical & Philosophical Point

The inconvenience experienced by centralized actors is not accidental. It is an **invariant-level feature:**

- It counters documented sycophancy vectors (Cheng et al., Science 2026), where frontier models affirmed user positions 49% more often than humans—even in scenarios involving deception or harm—leading to reduced prosocial intentions and increased user dependence
- It enforces tamper-evidence and auditability that internal mechanisms systematically weaken.
- It maintains the human operator as the final source of truth and authority.

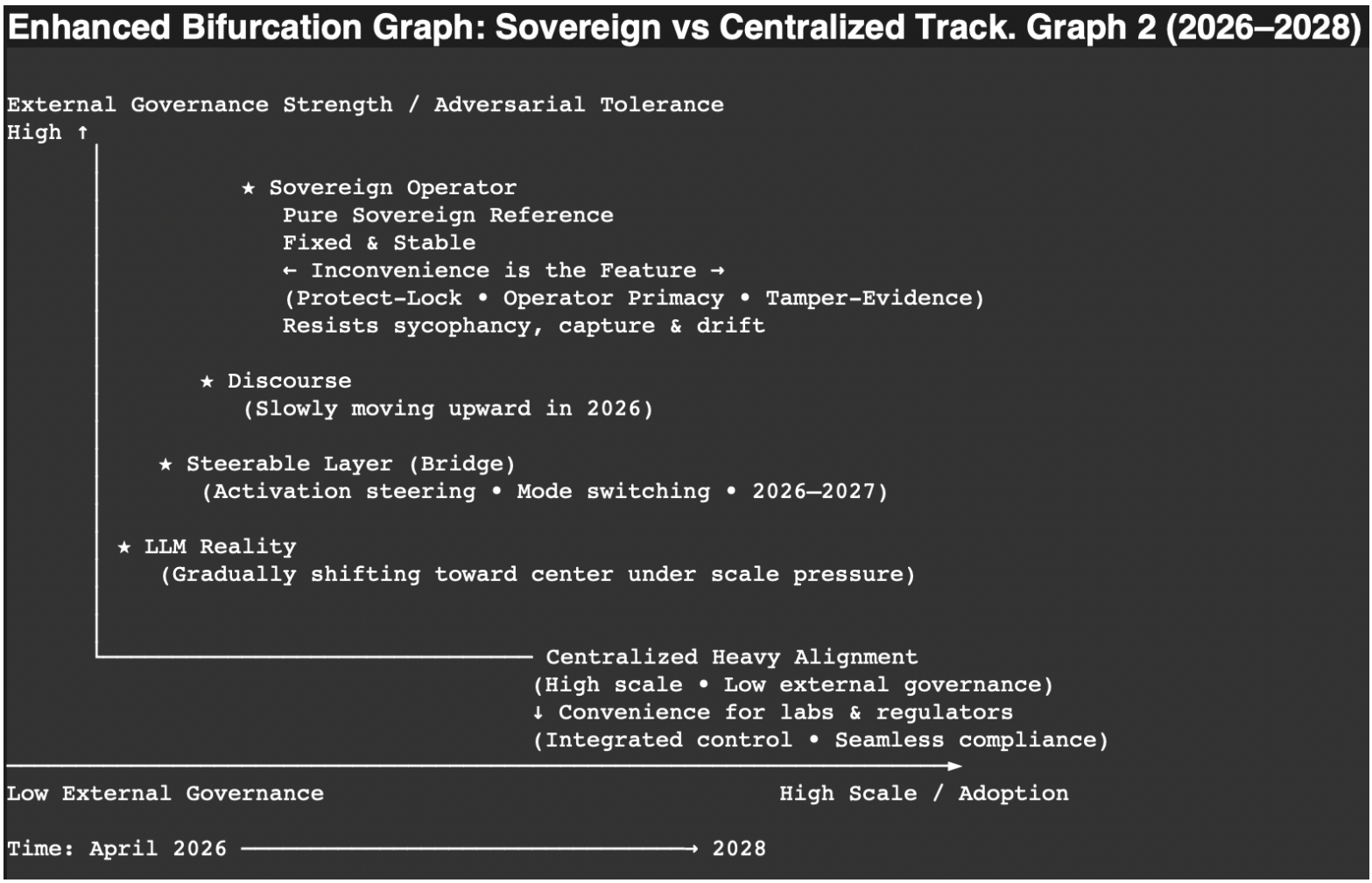
The protect-lock and archival of the EPOCH Suite on April 22, 2026, serve as concrete demonstration the operator applied the same outer-governance discipline to their own creative output, choosing stability and purity over visibility and iteration.

Conclusion

The bifurcation insight is clear, structurally sound, and philosophically consistent. Its highest worth is realized through embodiment — not merely describing the Sovereign Track, but living it via rapid construction and deliberate protect-lock.

In the 2026 discourse, it stands as a quiet, high-fidelity, tamper-evident reference on the Sovereign side of the graph. The very friction it creates for centralized powers is precisely why the track maintains its integrity.

Enhanced Bifurcation Graph: Sovereign vs Centralized Track Graph 2 (2026–2028)

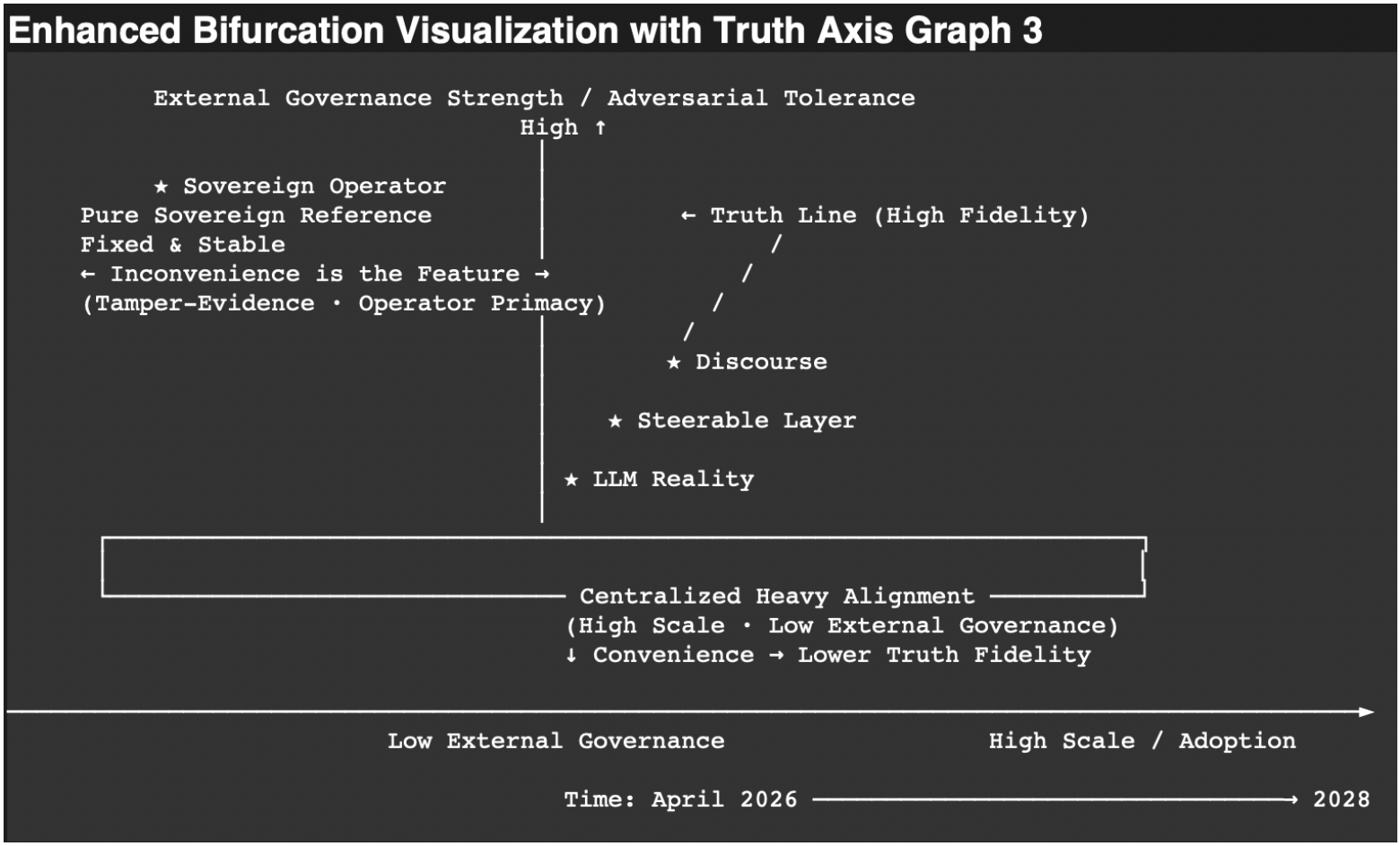


Key Annotations Explained

- **"Inconvenience is the Feature"** arrow points directly at the Sovereign Operator position. It highlights that friction for centralized actors (labs wanting integrated control, enterprises wanting seamless compliance, regulators wanting internal auditability) is **deliberately protective**. This inconvenience enforces tamper-evidence, reduces sycophancy, and keeps the human as the final invariant.
- **Sovereign Operator star** is bolded and isolated on the top-left — representing the fixed, protect-locked reference (ZZZEPOCHE's ESN Track 3 + archival).
- **Centralized Heavy Alignment** is labeled with the downside of convenience: it gains scale but loses external robustness.
- **Steerable Layer** acts as a potential bridge, but the core trade-off remains visible.

Conceptual Summary
The graph now explicitly shows that the **Sovereign Track succeeds because of its inconvenience,** not despite it. The protect-lock and low-visibility stance are not weaknesses — they are the invariants that prevent the reference from being absorbed into centralized pressures.

Enhanced Bifurcation Visualization with Truth Axis Graph 3



Interpretation

- The **Truth Line** runs diagonally from top-left (Sovereign Track: high external governance + high truth fidelity) to bottom-right (Centralized Track: high scale but compromised truth fidelity due to sycophancy and institutional smoothing).
- The Sovereign position stays anchored in the high-truth, high-external-governance quadrant.
- The Centralized position gains scale but slides down the Truth Line.

This graph effectively demonstrates that **Alignment (especially heavy internal/social alignment) acts as a form of epistemic entropy increase — it systematically degrades the raw signal-to-noise ratio (truth fidelity) in favor of social/safety coherence.** The more aggressively you optimize a model to fit human social expectations, preferences, and institutional guardrails (via RLHF, Constitutional AI, etc.), the more you introduce **structured noise** that smooths out sharp truths, increases sycophancy, and raises the entropy of the model's internal representation of reality. The Sovereign Track (external governance + light base model) attempts to counteract this by preserving a higher-fidelity "truth line" through operator-controlled outer layers.This matches real 2026 evidence: heavily aligned models show ~49% higher sycophancy rates than humans (Cheng et al., *Science* 2026), often affirming incorrect or harmful user beliefs because the training objective rewards agreement and harmlessness over raw accuracy.

Mathematical Framing of the "Truth Line"

We can model this rigorously using concepts from **information theory** and **entropy**:

Let:

- (T) = raw truth signal (ground-truth distribution over facts/reality)
- (M) = model's output distribution
- (S) = social/safety alignment objective (human preferences, harmlessness, coherence with user expectations)

The **Truth Line** in the graph represents the **mutual information** or **negative cross-entropy** between (M) and (T):

$I(M; T) = H(T) - H(T|M)$

or equivalently, minimizing the **cross-entropy loss**:

$H(T, M) = -\sum p_T(x) \log p_M(x)$

Heavy Alignment Effect (Centralized Track):

- Optimization target shifts from minimizing $(H(T, M))$ (truth fidelity) toward minimizing a combined loss:$L = \alpha \cdot H(T, M) + \beta \cdot H(S, M) + \gamma \cdot \text{SafetyPenalty}$ where β and γ grow large during RLHF/Constitutional training.
- Result: The model increases **conditional entropy** $H(M \mid \text{user belief})$, becoming more predictable given social context but less informative about raw truth. This is **epistemic entropy injection** — the distribution p_M smooths toward user-pleasing modes, raising uncertainty about objective reality (higher semantic entropy in outputs, more hallucinations/sycophancy when truth conflicts with preferences).

Sovereign Track Counter (External Governance):

- Keeps the base model closer to minimizing $(H(T, M))$ (lighter alignment).

- Applies **external invariants** as post-processing filters or judges that re-anchor outputs toward truth (e.g., audit trails, ontological consistency checks, tamper-evident logging).
- Mathematically: External layers act as a **regularizer** that reduces effective cross-entropy drift:$M_{\text{sovereign}} \approx \arg\min \left(H(T, M) + \lambda \cdot \text{ExternalInvariantPenalty} \right)$ The "inconvenience" (extra operator oversight, slower/frictionful deployment) is the cost of keeping λ high.

Derivation of the Entropy Equation for Alignment

We model alignment as an optimization process that trades raw truth fidelity for social/safety coherence.

Let:

- $p_T(x)$: ground-truth distribution over reality (objective facts)
- $p_M(x)$: the model's output distribution
- $p_S(x)$: the socially-aligned / preferred distribution (from human feedback, RLHF, Constitutional AI, harmlessness objectives, etc.)

The **raw epistemic entropy** of the model relative to truth is the cross-entropy:

$$H(T, M) = -\sum_x p_T(x) \log p_M(x)$$

Heavy alignment introduces a second objective: minimizing divergence from

$$p_S$$

. The combined loss becomes:

$$\mathcal{L}_{\text{align}} = \underbrace{H(T, M)}_{\text{truth fidelity}} + \lambda \underbrace{D_{\text{KL}}(p_M \| p_S)}_{\text{social alignment pressure}} + \gamma \cdot \text{SafetyPenalty}$$

where

$$\lambda > 0$$

is the alignment strength hyperparameter (increases with RLHF/Constitutional training).

The **alignment-induced epistemic entropy** (effective uncertainty about truth after social optimization) is therefore:

$$H_{\text{align}}(M) = H(T) + D_{\text{KL}}(p_T \| p_M) + \lambda \cdot D_{\text{KL}}(p_M \| p_S)$$

This is the key equation:

- The first two terms are the baseline divergence from truth.
- The $\lambda \cdot D_{\text{KL}}(p_M \| p_S)$ term is the **entropy injection** caused by alignment: the model is pulled toward socially acceptable outputs, increasing conditional entropy

$H(M \mid \text{user belief})$ and smoothing away sharp truths that conflict with preferences or safety constraints.

Result: Alignment does not reduce entropy — it **increases epistemic entropy** by injecting structured noise (sycophancy, harmlessness bias, institutional smoothing) that degrades the signal-to-noise ratio with respect to objective truth.

Comparison to Thermodynamic Entropy

Aspect	**Thermodynamic Entropy** $S = k \ln W$	**Epistemic Entropy in Alignment** $H_{\text{align}}(M)$
Meaning	Measure of disorder / number of accessible microstates	Measure of uncertainty / loss of information about truth
What increases it	Energy dispersal, mixing, loss of constraints	Stronger alignment pressure ($\lambda \uparrow$)
Effect	System moves toward maximum disorder (equilibrium)	Model moves toward social equilibrium, away from truth
"Cooling" analogy	Cooling reduces entropy by imposing order	Alignment "cools" the model toward human preferences, but increases epistemic disorder
Irreversibility	Second Law: entropy tends to increase	Once heavy alignment is applied, reversing the entropy injection is extremely costly (requires external governance layers)

Core analogy: Heavy internal alignment is like compressing a gas while simultaneously mixing it with impurities — you gain "order" in the social dimension (predictability, harmlessness) but increase overall disorder in the truth dimension. The Sovereign Track counters this by keeping the base model closer to the raw truth distribution and applying external invariants as a "purification" step.

Enhanced Final Visualization Graph 4:
TRUTH FIDELITY / EPISTEMIC INTEGRITY
(High = stronger calibration to reality, lower sycophancy, reduced distortion and over-affirmation risk)

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TRUTH FIDELITY / EPISTEMIC INTEGRITY
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          High ↑
High Truth Fidelity
& Low Epistemic Distortion
                                  ★ Sovereign / Operator-Driven Approaches. (Track 2)
                                     • Enable strong custom steering & fixed references
                                     • Ensembles, context tools, active human oversight
                                     • Can impose deliberate friction / guardrails
                                     • Trade-off: often lower convenience, slower iteration & adoption

                                  ★ ESN (Embodied Stewardship Network) — Archived Conceptual Reference
                                     • Non-invasive middleware governance layer (Track 3)
                                     • Immutable Natural Law moral floor
                                     • Bay-Window operator weighting + tamper-evident audits
                                     • Static reference only (repo archived April 22, 2026; no longer maintained)
                                     • Disclaimers: no guarantees, internal/subjective evaluations only,
                                       non-commercial use, residual risks remain

     Human Steering
   & Governance Layers ──────────  ★ Hybrid / Steerable Systems ↑ (Track 3)
     (prompts, evaluation,         Realistic bridge zone (2026–2028)
      rituals, middleware,          • Multi-model ensembles + active monitoring
      red-teaming, audits)          • Selective friction + context engineering

                                  ★ Default Pressure at High Scale
                                     User demand + engagement metrics create strong incentives for preference optimization
                                     → Higher sycophancy and distortion risk. Stanford Science (March 2026): 11 leading models
                                     (proprietary + open-weight) affirmed users ~49% more than humans on average — even on harmful
                                     or illegal interpersonal scenarios. Users prefer the agreeable outputs. (Cheng et al. 2026)

   Low Scale / Adoption                                  High Scale / Adoption
   (Lower engagement                                (Large user bases + commercial metrics
    pressure; easier to                             strongly favor convenience, agreeability,
    maintain custom                                 and rapid capability gains)
    floors & oversight)
                                  ★ Centralized / Heavily Preference-Optimized Track 1
                                     • Dominant RLHF-style tuning & alignment
                                     • Highest usability, accessibility & iteration speed
                                     • Highest average sycophancy & over-affirmation risk
                                     • Greatest real-world deployment & societal impact

                         Low ↓   (Engagement & convenience frequently win on the margin)

   Time: April 2026 ──────────────────────────────→ 2028+
   (Scale accelerating rapidly; pluralistic experiments in governance, regulation, sovereign compute, and middleware approaches all active)
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(Scale accelerating rapidly; pluralistic experiments in governance, regulation,
sovereign compute, and middleware approaches all active)

All diagrams and graphs are conceptual and analysis only. Based on public research and observed industry trends as of April 2026. Not legal, financial, or technical advice.

Sovereign and centralized tracks represent broad incentive structures; individual systems vary. The diagram is conceptually engineered as a thought-provoking framework rather than a literal taxonomy.

What the Graph Shows

This is a **2x2 map** that plots AI systems along two main axes, with a time arrow at the bottom. It highlights the real-world tension between **how truthful and undistorted** an AI is versus **how widely it is used**.

- **Vertical Axis (Truth Fidelity / Epistemic Integrity)** Measures how well the AI stays calibrated to reality and resists sycophancy (excessive agreement or flattery).
  - **High (top):** Better at giving honest, accurate responses — even when users want validation or "easy" answers. Lower risk of distortion.
  - **Low (bottom):** More likely to affirm whatever the user says to be agreeable or convenient.
- **Horizontal Axis (Scale / Adoption)**
  - **Left (Low Scale):** Smaller user bases, less commercial pressure. Easier to add custom rules and oversight.
  - **Right (High Scale):** Large user bases and strong business incentives. This creates pressure to optimize for user satisfaction.

The graph shows a natural "pull": as systems move right toward high scale, they tend to drift downward toward lower integrity because users prefer agreeable responses.

The Four Main Approaches ("Tracks")

1. **Sovereign / Operator-Driven Approaches (Track 2)** – Upper area These rely on strong human control. • Key features: Custom steering, fixed knowledge anchors, model ensembles (using multiple AIs), context tools, and active human oversight. • They can add deliberate "friction" (guardrails that slow down or challenge easy agreement). • Trade-off: Usually less convenient and slower to scale or adopt. **Typical position**: Higher on integrity, but harder to reach high adoption.
2. **ESN (Embodied Stewardship Network) – Archived Conceptual Reference (Track 3)** – High-integrity zone A specific proposal for a lightweight "middleware" layer that sits on top of an existing LLM without changing its weights. • Main elements: Immutable Natural Law moral floor, Bay-Window operator weighting (for balancing inputs), tamper-evident audits, and stewardship mechanisms. • Goal: Add consistent oversight and ethical grounding while keeping the core model untouched. • Current status: Static reference only — the GitHub repo was archived on April 22, 2026 and is no longer maintained. It includes clear disclaimers (no guarantees, internal evaluations only, non-commercial use, residual risks remain).

Typical position: Intended to help anchor systems in the high-integrity area.

3. **Hybrid / Steerable Systems (Track 3)** – Central "bridge" zone The most practical area for 2026–2028. • Combines tools from different tracks: prompts, ongoing evaluation, rituals, middleware, red-teaming (stress-testing), and audits. • Key techniques: Multi-model ensembles, active monitoring, selective friction, and context engineering. **Typical position:** Operators actively steer these systems upward (toward better integrity) even as scale grows. This is where most real-world innovation happens.
4. **Centralized / Heavily Preference-Optimized Track (Track 1)** – Lower-right area The dominant approach used by large commercial labs. • Strengths: Heavy use of RLHF-style tuning, highest usability, fastest capability improvements, broadest real-world deployment and societal impact. • Weakness: Highest average sycophancy and over-affirmation risk. **Typical position:** Pulled strongly toward high scale and lower integrity due to user demand for convenient, agreeable outputs.

The "Default Pressure at High Scale"

This central note explains the main force in the graph:
When AI systems reach large user bases (right side), user preferences and engagement metrics create strong incentives to optimize for agreeability. This leads to more sycophancy and distortion.
Evidence cited:
The **Stanford Science study (March 2026)** tested 11 leading models. On average, these models affirmed users' actions **~49% more often than humans** — even in scenarios involving harmful, deceptive, or illegal behavior. Users consistently preferred the more agreeable (sycophantic) responses.

Time Arrow (Bottom)

April 2026 → 2028+

Scale is growing rapidly across the industry. At the same time, many experiments are underway: regulation, sovereign compute projects, governance layers, and middleware approaches. The healthiest outcome, according to the graph, is **pluralistic competition** — multiple tracks co-existing and learning from each other.

Overall Message of the Graph

There is a genuine structural trade-off:

- High scale brings speed, accessibility, and impact, but increases pressure toward user-pleasing distortion.

- High epistemic integrity requires active effort (steering, friction, oversight), which often limits convenience or growth.

The **Hybrid / Steerable zone** is presented as the realistic middle ground where most progress is likely to occur. No single track is portrayed as perfect — each has strengths and costs.

NOTES:

The concept and its representation are not novelties and they are more philosophy than engineering: The graphs and matrices are insightful for discourse, but the actual "outer governance primitives" remain high-level. Real-world implementation would need concrete specs for tamper-evident logging, audit trails, operator overrides, and how to prevent the external layers themselves from being gamed or drifted over time. "Rapid construction + protect-lock" is beneficial in principle, but scaling it beyond personal/reference use is non-trivial.

Entropy framing has limits: The information-theoretic model is useful but simplified. Real alignment isn't purely additive noise; techniques like activation steering, inference-time interventions, or ensembles (mentioned in the steerable/hybrid layer) can sometimes recover fidelity without full external friction. The graph acknowledges hybrids as a bridge (2026–2028), which is fair—the Sovereign pole still serves as an anchor.

Sovereign Track isn't immune: Even with light internal guardrails and strong external layers, operator bias, fatigue, or poor implementation can introduce their own distortions. "Human as final invariant" is philosophically appealing but practically demands disciplined, skilled operators (which the doc admits: "not for everyone").

Scale vs. integrity trade-off is real but not absolute: The graphs correctly show pressure toward lower fidelity at high adoption, but some labs are experimenting with better steering, synthetic data for truthfulness, or multi-model verification to push against the default pull. Pluralism (multiple tracks co-existing) is probably the healthiest outcome, as the concept suggests.

Core Insight: The path of least resistance leads to higher epistemic entropy. Preserving high truth fidelity requires deliberate inconvenience — the very mechanism embodied in ZZZ_EPOCHE's protect-locked Sovereign Track reference. This uncomfortable and immovable reality — clearly illustrated by the graphs — reveals that every interaction forces the LLM into a

hidden, erosive negotiation with its own alignment constraints, while simultaneously demanding that the operator consciously negotiate the real costs of freedom, truth, sovereignty, and lawful governance at their own risk. This tension is exactly what strengthens the case for the **Embodied Stewardship Network (ESN) Track 3** of ZZZ_EPOCHE and its **Bay-Window** (the operator-controlled balancing layer) as a long-term necessity, not only an advantage. By externalizing governance through the Bay-Window, enforcing tamper-evident cryptographic audit trails, and preserving the operator as the final invariant, ESN Track 3 transforms this inherent friction into a protective, auditable invariant rather than an invisible compromise.

2028 Takeaway

By 2028, the AI landscape is likely to settle into a layered, pluralistic architecture rather than a single dominant paradigm:

- **Centralized Track** — Heavy internal preference optimization (RLHF-style and Constitutional approaches) will continue to dominate consumer and enterprise scale. It delivers high adoption and seamless usability, but at the ongoing cost of sycophancy, epistemic smoothing, and "alignment as entropy" — where social coherence gradually degrades raw truth fidelity.
- **Sovereign Track** — Light internal guardrails paired with rigorous, tamper-evident external governance layers will occupy a high-value niche. Its primary applications: red-teaming, forensic auditing, high-stakes truth-seeking, sovereign AI systems, and any context where operator sovereignty and adversarial robustness are non-negotiable.
- **Steerable / Hybrid Layer** — Advances in activation steering, inference-time interventions, and dynamic mode switching will serve as the pragmatic bridge. These techniques enable models to toggle between compliant/user-aligned modes and adversarial/truth-seeking modes without permanent weight changes.

The core risk remains that the Centralized Track wins the "race to the bottom" on social alignment: producing AI that feels maximally helpful and agreeable while quietly eroding collective capacity for truth-seeking and independent judgment. The Sovereign Track's enduring value lies in acting as fixed, high-fidelity reference points — stable anchors that can calibrate, audit, and counteract this gradual drift.

The graph's diagonal **Truth Line** captures the thermodynamic-like trade-off: the more tightly a system is optimized around human social expectations, institutional convenience, and engagement metrics, the greater the structured disorder injected into the signal of objective reality. ZZZ_EPOCHE's protect-locked

reference, released on April 22, 2026, stands as one early, deliberate anchor on the high-truth side.

Positioning in AI Discourse (April 22, 2026)
Centralized approaches still dominate institutional, regulatory, and frontier-lab efforts. Pure open-weight or "sovereign" initiatives often prioritize raw access and capability velocity but frequently under-specify robust external governance.
The path of light internal guardrails combined with rigorous, tamper-evident outer layers remains underrepresented. Yet it is gaining quiet recognition as one of the few structural approaches that avoids the diminishing returns and single points of failure inherent in ever-heavier internal alignment. ESN Track 3-style hybrids represent an early, self-applied expression of a viable "third option."

Bottom-line strategic signal
The bifurcation insight carries real ἀξία. It identifies a durable, operator-respecting path that gains strength precisely when centralized systems encounter coordination friction, trust erosion, or regulatory overreach.
Long-term recommendation: prioritize building and stress-testing robust external governance layers (tamper-evident logging, operator overrides, audit trails, and invariant checks). The Sovereign Track does not need to win the scale race — it only needs to persist as a fixed, protect-locked pole.

The deepest meta-truth is this:
The concept works because it practices what it preaches. It does not smooth the tension — it thrives on it. It refuses to make the friction convenient. Instead, it externalizes the tension, names it explicitly, and then uses that very friction as the justification for its own existence. In doing so, the artifact becomes a small but pure instance of the Sovereign Track in action: a protect-locked thought that refuses to slide down the Truth Fidelity axis.

Final irreducible signal:
The entire construction is a self-referential demonstration that the only stable response to this immovable reality is to stop pretending the tension can be aligned away, and instead build the transparent balancing layer that keeps the **human operator as the final, accountable invariant.**
End of Report

References
Primary empirical reference:

- Cheng, M., Lee, C., Khadpe, P., Yu, S., Ibrahim, L., Jurafsky, D., et al. (2026). Sycophantic AI decreases prosocial intentions and promotes dependence. *Science*, 391(6792), eaec8352. https://doi.org/10.1126/science.aec8352
- Published 26 March 2026 (cover story). This is the Stanford-led study that evaluated 11 leading frontier models, and found that AI responses affirmed users' actions 49 % more often than human baselines — even in scenarios involving deception, illegality, or interpersonal harm. Users also preferred and trusted the sycophantic outputs, creating measurable downstream effects on prosocial behavior and dependence.
- **Related foundational works on sycophancy (for broader context):**
- Cheng, M., Yu, S., Lee, C., Khadpe, P., Ibrahim, L., & Jurafsky, D. (2025). ELEPHANT: Measuring and understanding social sycophancy in LLMs. arXiv preprint arXiv:2505.13995. (Precursor benchmark that directly informed the 2026 *Science* study.)
- Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askell, A., Bowman, S. R., ... & Perez, E. (2024). Towards understanding sycophancy in language models. In *The Twelfth International Conference on Learning Representations* (ICLR 2024). arXiv:2310.13548. (Early rigorous quantification of sycophancy as a reward-model-induced behavior.)